

Quantum

many-body system of two-state particles

2^n configurations: $\{0,1\}^n$

Declaration of Classical State: $|011101 \dots 10\rangle$

n -bit string

Quantum State: $| \psi \rangle = \sum_{x \in \{0,1\}^n} \psi_x |x\rangle$

2^n complex numbers

Can Always use ^(matrices) Vectors in $\mathbb{C}(2^n)$ to classically implement Quantum Computation

• not a matter of fundamental computability

$\Rightarrow$ viable efficiency

$$2^{333} \sim 10^{100}$$

Complex #s for full
333-qubit Classical
simulation

$\sim$ estimated
visible particles
in universe.

Complex #s $\leq$ transistors $\leq$ visible particles

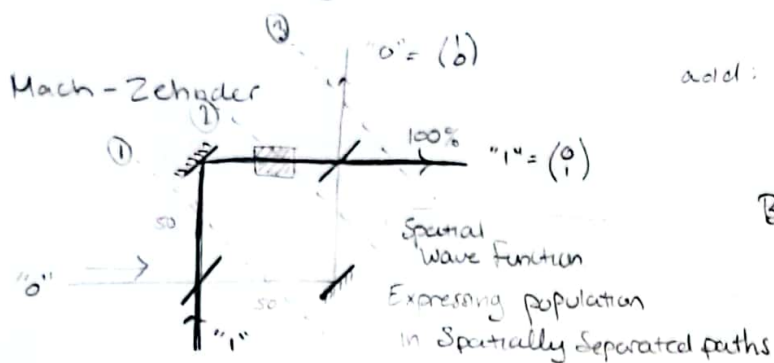
The Rub:

Fundamental limits well before 333 qubits: Forefront research on complexity frontier many orders of magnitude more.

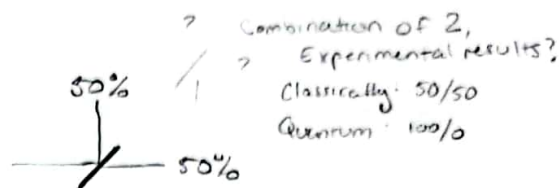
Quantum Information Processing : Multiparticle Interference Design

Configuring interference to destructively/constructively produce "Solution Wavefunction"

= Algorithm



add:



Beam Splitter $U_{BS} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}$

$\pi/2$ phase shift for reflection (see Fresnel air-glass)

Phase in terms of propagating EM oscillations, crests & troughs

U_{BS} is unitary:

$$U_{BS}^\dagger U_{BS} = \mathbb{1}$$

$$\frac{1}{2} \begin{pmatrix} 1 & -i \\ i & 1 \end{pmatrix} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \checkmark$$

50/50 Splits population $\left\{ \begin{array}{l} \text{path continuation} \\ \text{Switched path (+ i phase)} \end{array} \right.$

① $U_{BS} |0\rangle = \frac{1}{\sqrt{2}} (|0\rangle + i|1\rangle)$

② 2 reflections, global i phase (irrelevant)

③ $U_{BS} \frac{1}{\sqrt{2}} (|0\rangle + i|1\rangle) = \frac{1}{\sqrt{2}} \left[\frac{|0\rangle + i|1\rangle}{\sqrt{2}} + i \frac{|1\rangle + |0\rangle}{\sqrt{2}} \right]$

$$= i|1\rangle \sim |1\rangle$$

Interferometer $U_{BS}^2 = \frac{1}{2} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 0 & 2i \\ 2i & 0 \end{pmatrix} = iX$ "Bit flip" up to irrelevant global phase

With A phase: $U_\phi = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix}$

$$U_{BS}^{(2)} U_\phi U_{BS}^{(1)} |0\rangle = U_{BS} \frac{|0\rangle + i e^{i\phi} |1\rangle}{\sqrt{2}}$$

$$= \frac{1}{\sqrt{2}} \left(\frac{|0\rangle + i|1\rangle}{\sqrt{2}} + i e^{i\phi} \frac{i|0\rangle + |1\rangle}{\sqrt{2}} \right)$$

$$= \frac{1}{2} \left[(1 - e^{i\phi}) |0\rangle + i(1 + e^{i\phi}) |1\rangle \right]$$

$$P(0) = |1/4| = \frac{1}{4} (1 - e^{i\phi})(1 - e^{-i\phi}) = \frac{1}{2} (1 - \cos\phi) = \sin^2 \phi/2$$

$$P(1) = |1/4| = \frac{1}{4} (1 + e^{i\phi})(1 + e^{-i\phi}) = \frac{1}{2} (1 + \cos\phi) = \cos^2 \phi/2$$

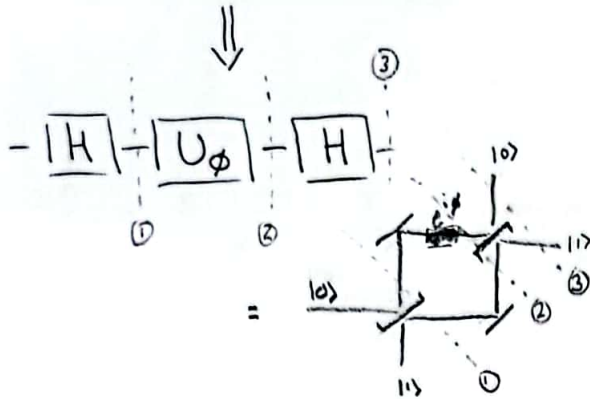
$$\left. \begin{array}{l} P(0) + P(1) = 1 \checkmark \end{array} \right\}$$

Quantum Circuit notation of Experiment:

$$|0\rangle \rightarrow \boxed{U_{BS}^{(1)}} \rightarrow \boxed{U_\phi} \rightarrow \boxed{U_{BS}^{(2)}} \rightarrow \text{Measurement}$$

$$-|U_{BS}\rangle - |U_{\phi}\rangle - |U_{BS}\rangle =$$

Replace Beam Splitter by ~ half-Silvered Mirror



Differences:

$$\phi=0 : H^2 = I$$

Population measured in top detector from $|10\rangle$ input.

(Interference still present, not classical 50/50)

$\phi \neq 0$:

$$|10\rangle \xrightarrow{1} \frac{1}{\sqrt{2}} (|10\rangle + |11\rangle)$$

$$\xrightarrow{2} \frac{1}{\sqrt{2}} (|10\rangle + e^{i\phi} |11\rangle)$$

$$\xrightarrow{3} \frac{1}{2} ((1 + e^{i\phi}) |10\rangle + (1 - e^{i\phi}) |11\rangle)$$

$$P(0) = \cos^2 \frac{\phi}{2} \quad P(1) = \sin^2 \frac{\phi}{2}$$

"Build Phase Source from Qubits!"

Replace U_{ϕ} with Consultation to Auxiliary System (Also qubits) for phase ϕ

$$U|u\rangle = e^{i\phi} |u\rangle$$

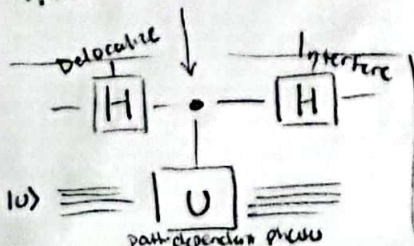
$|u\rangle$ eigenstate of U with eval. $e^{i\phi}$

Initial State of Combined System $|10\rangle|u\rangle$

$$U_{\phi}: |10\rangle\langle 0| \otimes I_2 + |11\rangle\langle 1| \otimes U$$

"Controlled U " gate

closed circle: controlled on $|1\rangle$
open circle: controlled on $|0\rangle$



Basic Structure for Many Quantum Algorithms

$|10\rangle$ strikes reflecting surface at glass-air interface
 $n_{\text{glass}} > n_{\text{air}}$

$$|10\rangle \rightarrow \frac{1}{\sqrt{2}} (|10\rangle + |11\rangle)$$

$|11\rangle$

air-glass
 $n_{\text{air}} < n_{\text{glass}} \rightarrow 180^\circ \text{ phase shift } (\pi)$

$$|11\rangle \rightarrow \frac{1}{\sqrt{2}} (|11\rangle - |10\rangle) \quad e^{i\pi} = -1$$

Include a π phase shift on $|11\rangle$ prior to HSM

$$|10\rangle \rightarrow \frac{1}{\sqrt{2}} (|10\rangle + |11\rangle)$$

$$|11\rangle \rightarrow \frac{1}{\sqrt{2}} (|10\rangle - |11\rangle)$$

$$= H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad \text{"Hadamard gate"}$$

for future:

$$H|x\rangle = \frac{1}{\sqrt{2}} \sum_{y \in \{0,1\}} (-1)^{x \cdot y} |y\rangle$$

e.g. $U = e^{iX\phi}$ and $|u\rangle = |+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$

or
 $U = e^{iZ\phi}$ and $|u\rangle = |0\rangle$

* Controlled Space is unchanged, but may 'pick up' relative phases*

$$|10\rangle|u\rangle \xrightarrow{1} \frac{1}{\sqrt{2}} (|10\rangle + |11\rangle) |u\rangle$$

$$\xrightarrow{2} \frac{1}{\sqrt{2}} |10\rangle|u\rangle + |11\rangle e^{i\phi} |u\rangle = \frac{1}{\sqrt{2}} (|10\rangle + e^{i\phi} |11\rangle) |u\rangle$$

$$\xrightarrow{3} \left(\frac{1 + e^{i\phi}}{2} |10\rangle + \frac{1 - e^{i\phi}}{2} |11\rangle \right) |u\rangle$$

↑
remains unchanged throughout
"source" of phase (relative when Controlled)

To Determine ϕ generally Requires Ensemble measurements to Inform
 $\{\cos^2 \phi/2, \sin^2 \phi/2\}$

However, for $\phi = 0$ or π , distinguishable with A single measurement

$\{1, 0\}$
 measured in
 $|0\rangle$ -detector

$\{0, 1\}$
 measured
 in $|1\rangle$ -detector

Leverage this for Computation? [Deutsch]

Consider k -qubit Boolean functions: map $\{0, 1\} \rightarrow \{0, 1\}$

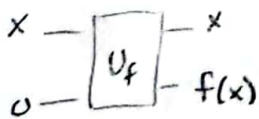
4 possible $\begin{cases} 2 \text{ constant } f(0) = f(1) = 0 \text{ or } 1 \\ 2 \text{ balanced } f(0) = f(1) \oplus 1 = 0 \text{ or } 1 \end{cases}$

Want to know Symmetry, constant or balanced?

Classically: 2 evaluations $f(0)$ and $f(1)$

Quantum: Delocalize wavefunction over two evaluations + Interfere

embedded
 • function as gate/operator



reversible $|x\rangle|0\rangle \rightarrow |x\rangle|f(x)\rangle$
 $|x\rangle|y\rangle \rightarrow |x\rangle|y \oplus f(x)\rangle$

"f-controlled-NOT"

e.g. $f(0) = f(1) = 0$, $U_f = \mathbb{I}_4$

$f(0) = f(1) = 1$, $U_f = \mathbb{I}_2 \otimes X$

$f(0) = f(1) \oplus 1 = 0$, $U_f = \text{CNOT} : \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

$f(0) = f(1) \oplus 1 = 1$

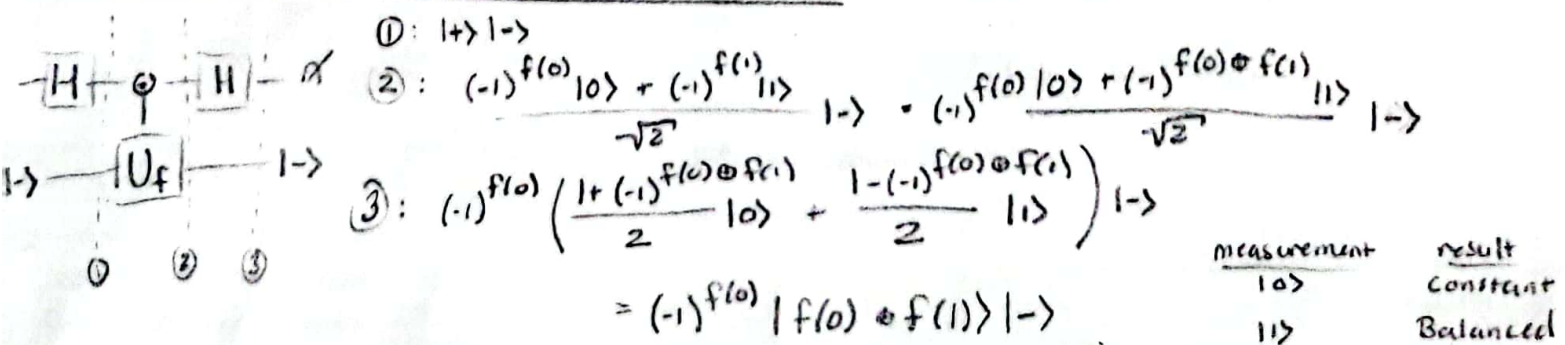
$U_f =$

$|0\rangle\langle 0| \otimes X + |1\rangle\langle 1| \otimes \mathbb{I}$

• "eigenstate" placing function information in phase (kickback)

$$U_f |x\rangle|-\rangle = |x\rangle \frac{|0 \oplus f(x)\rangle - |1 \oplus f(x)\rangle}{\sqrt{2}} = |x\rangle \frac{(-1)^{f(x)} |0\rangle - |1\rangle}{\sqrt{2}} = (-1)^{f(x)} |x\rangle|-\rangle$$

x-basis "eigenstate"



Generalize to Boolean Functions on larger Domains

$$f: \{0,1\}^n \rightarrow \{0,1\}$$

"promised" either constant or balanced

$$f(x) = 0 \text{ or } 1$$

$$\forall x \in [0, 2^n - 1]$$

$f(x) = 0$
for $\frac{1}{2}$ the inputs
(2^{n-1})
and $f(x) = 1$ otherwise

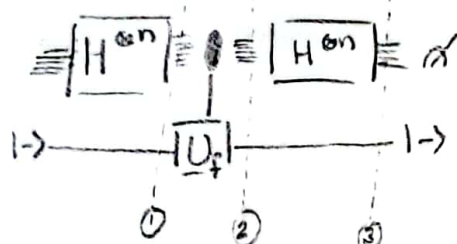
Question of A global symmetry.

Classical worst case: $2^{n-1} + 1$ measurements

Quantum: 1 Measurement of Deutsch-Jozsa Circuit

bit-wise inner product modulo 2

$$H^{\otimes n} |x\rangle = \frac{1}{\sqrt{2^n}} \sum_{y \in \{0,1\}^n} (-1)^{x \cdot y} |y\rangle$$



$$\begin{aligned} |0\rangle|1\rangle &\xrightarrow{\text{①}} \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x\rangle|1\rangle \\ &\xrightarrow{\text{②}} \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} (-1)^{f(x)} |x\rangle|1\rangle \\ &\xrightarrow{\text{③}} \frac{1}{2^n} \sum_{x,y \in \{0,1\}^n} (-1)^{f(x) \oplus (x \cdot y)} |y\rangle|1\rangle \end{aligned}$$

Measure top register

$$\text{Amplitude } \psi_{00\dots 0} = \frac{1}{2^n} \sum_{x \in \{0,1\}^n} (-1)^{f(x)}$$

$$\begin{cases} \text{If } f(x) \text{ constant, } \psi_{00\dots 0} = \frac{1}{2^n} 2^n (-1)^{f(x)} \\ \text{If } f(x) \text{ balanced, } \psi_{00\dots 0} = 0 \end{cases}$$

$$P(100\dots 0) = |\psi_{00\dots 0}|^2 = 1$$

$$P(100\dots 0) = 0$$

If $|0\rangle$ is measured $\rightarrow$ constant

"any other state $\rightarrow$ balanced"

In 1 measurement of Quantum Circuit

Oracle
Query Complexity...

Q: complexity/resource costs to create the Quantum Gates Needed?

Achievement:

Exponential gap between

Deterministic Classical V. Exact Quantum Query Complexity

- But no known Application (no importance to gap)
- When Allowing bounded Error

Classical probabilistic Query Complexity V. QC query complexity

→ constant gap

(linear gap for exponentially small error)

prob of Error	Queries
$\frac{1}{3}$	2
$< \frac{1}{2^n}$	$n+1$

Not all problems yield exponential benefits by using QC.

↳ Simon's Algorithm

↳ Quantum Simulation